Data source

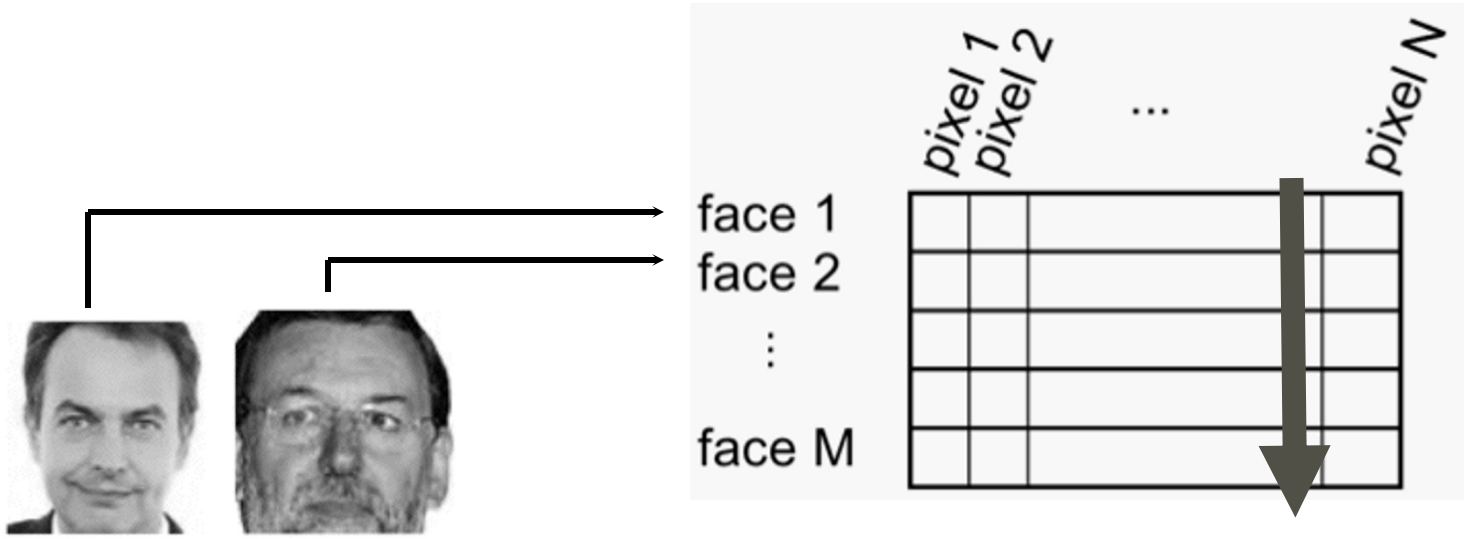
- Creation of the dataset
  - Formulated as a web data mining problem
  - We coded a script in `phyton` to extract infor
    Representatives:
    - http://www.congreso.es



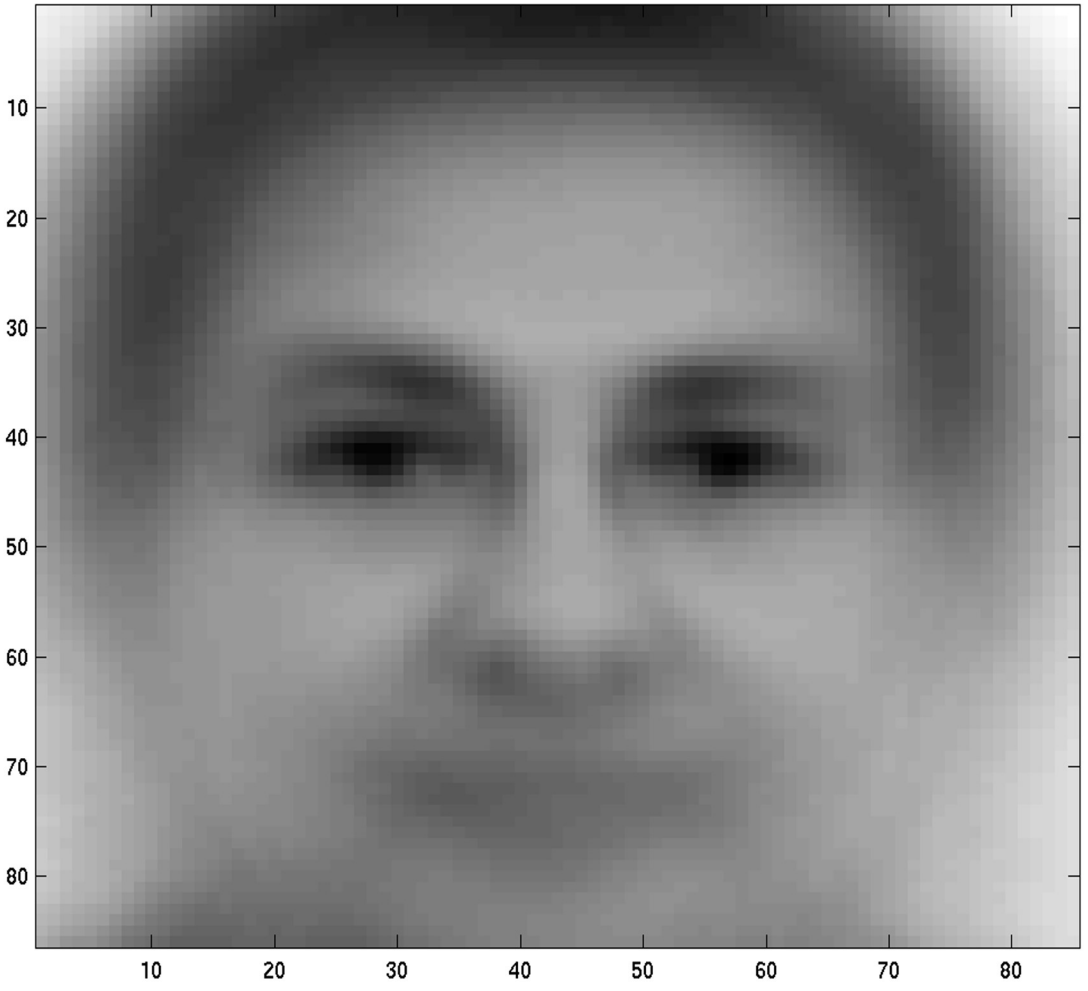

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Data Matrix construction

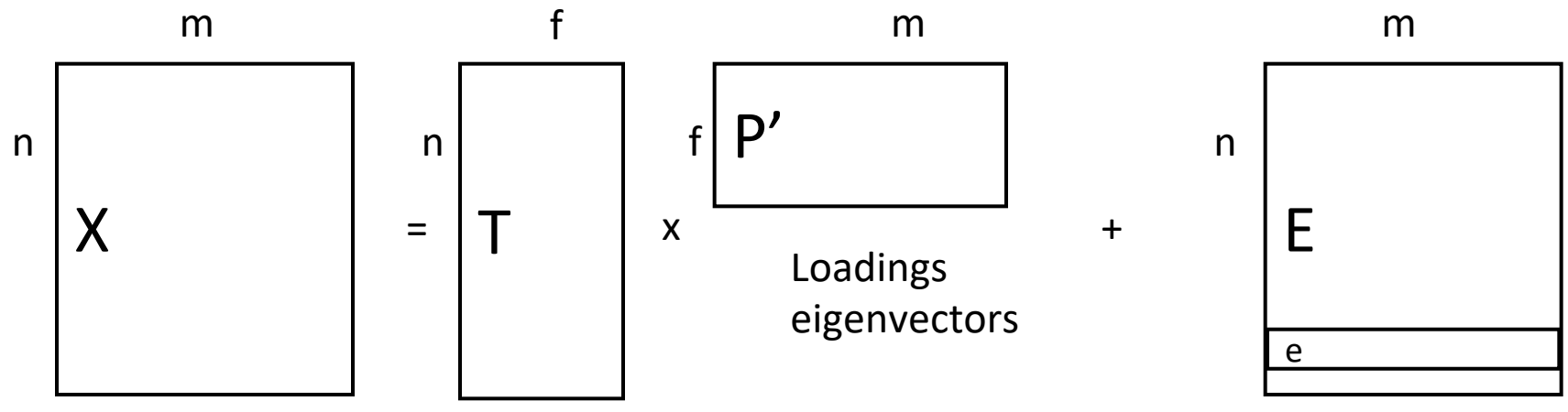
- Dataset analysis
  - Images converted to Black and white
    - 86 x 85 = 7310 features (pixels)
    - 350 patterns corresponding to 350 representatives

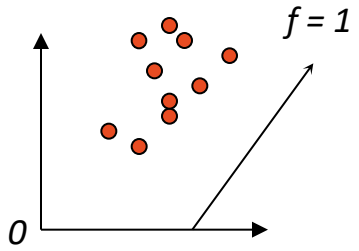


Eigenfaces

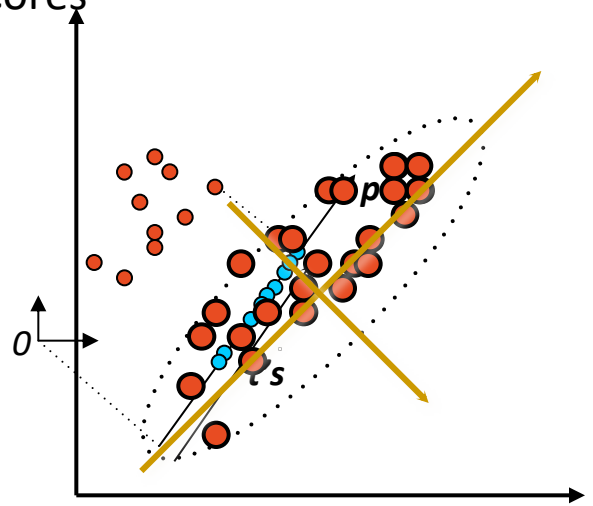


Karhünen Loéve transformation

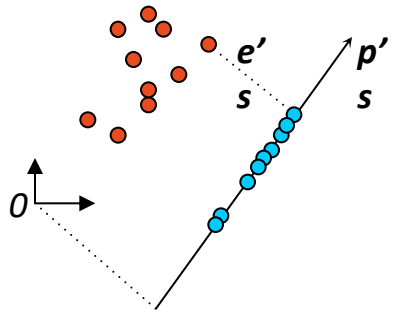




m = 2

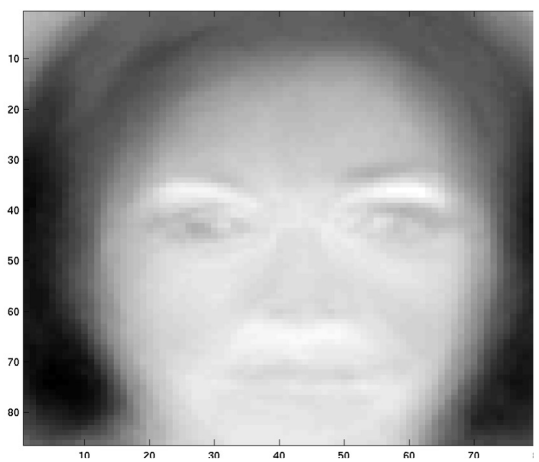


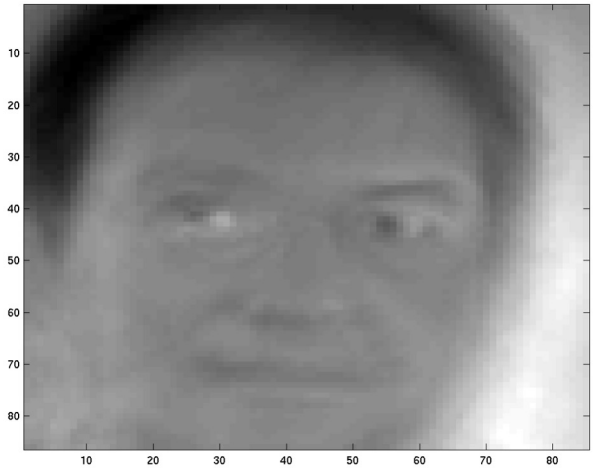
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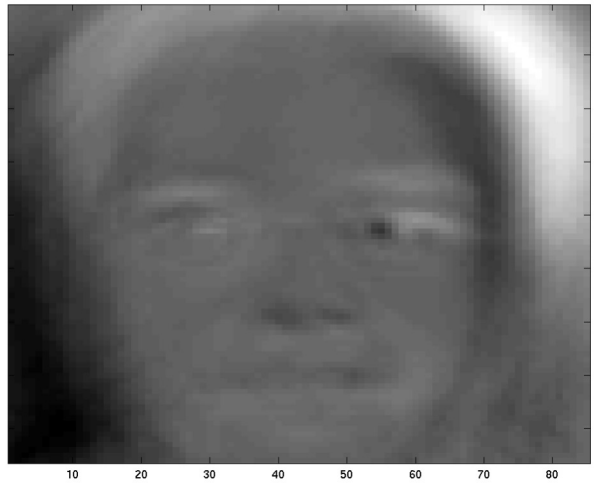


Eigenfaces


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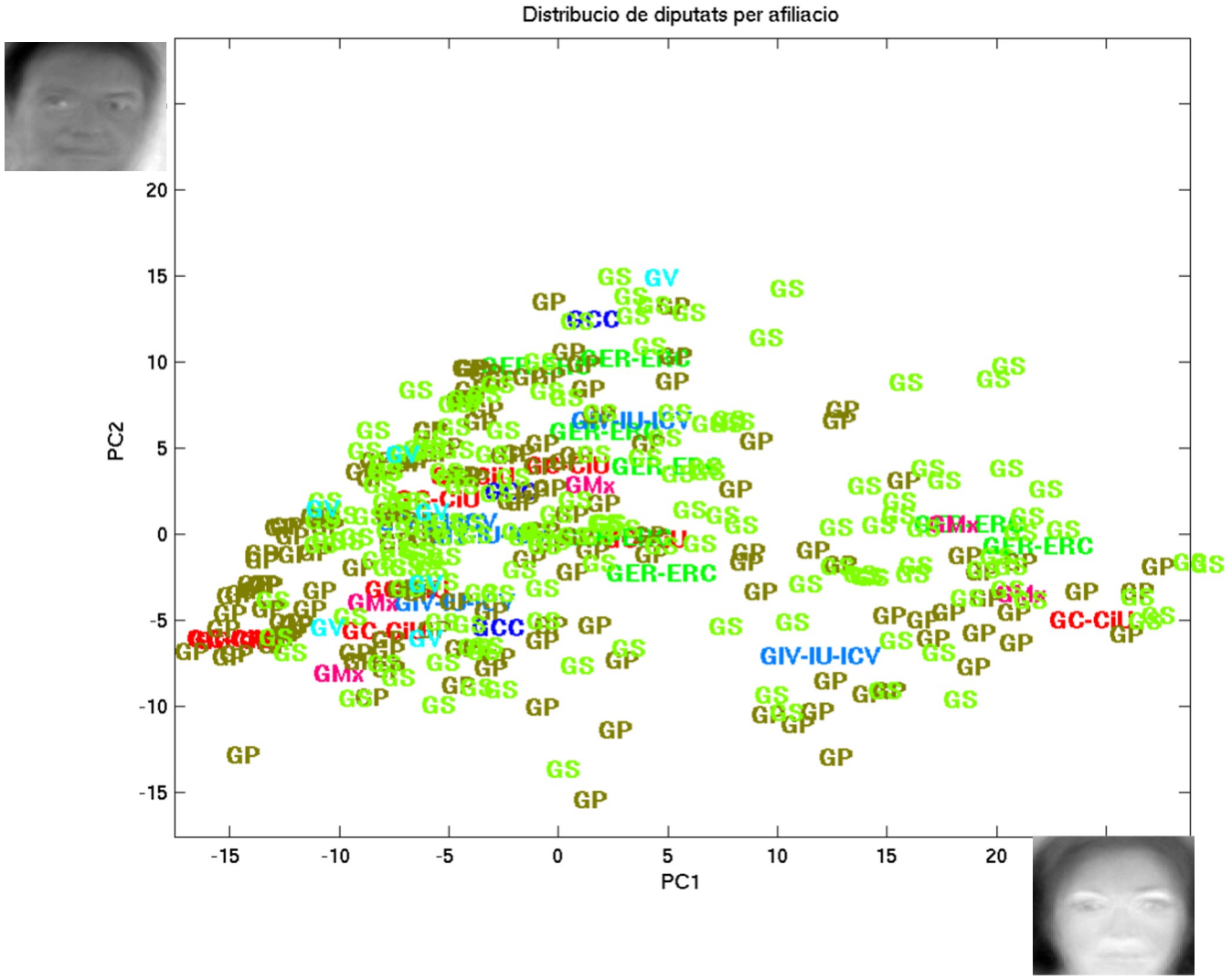

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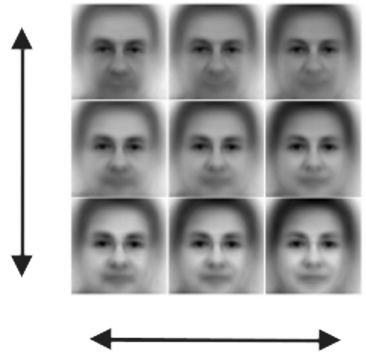

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PCA score plots

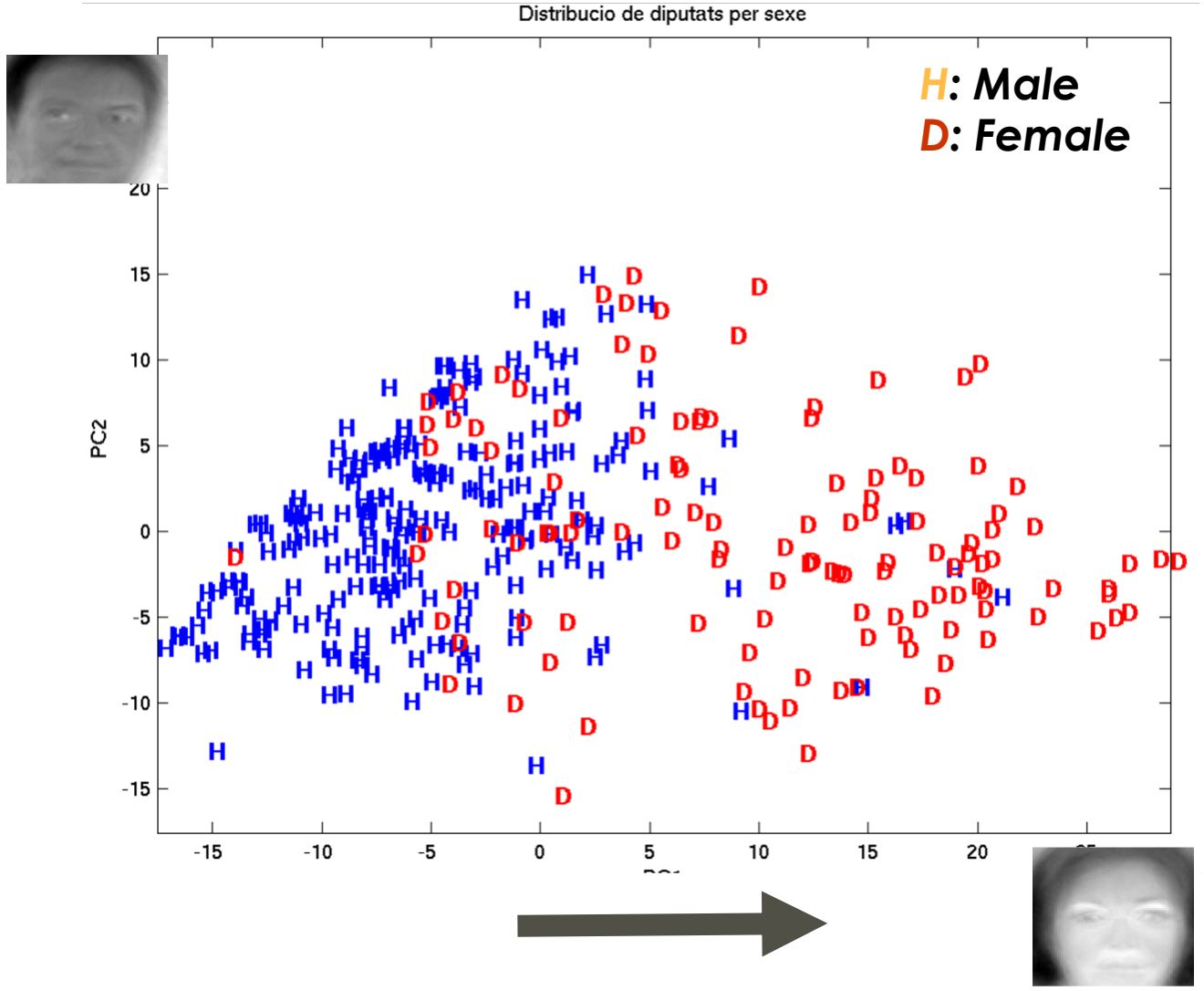
- PCA plot, Main variance not correlated with affiliation

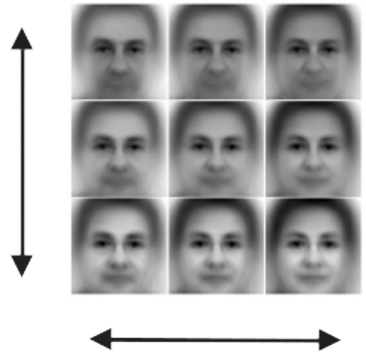


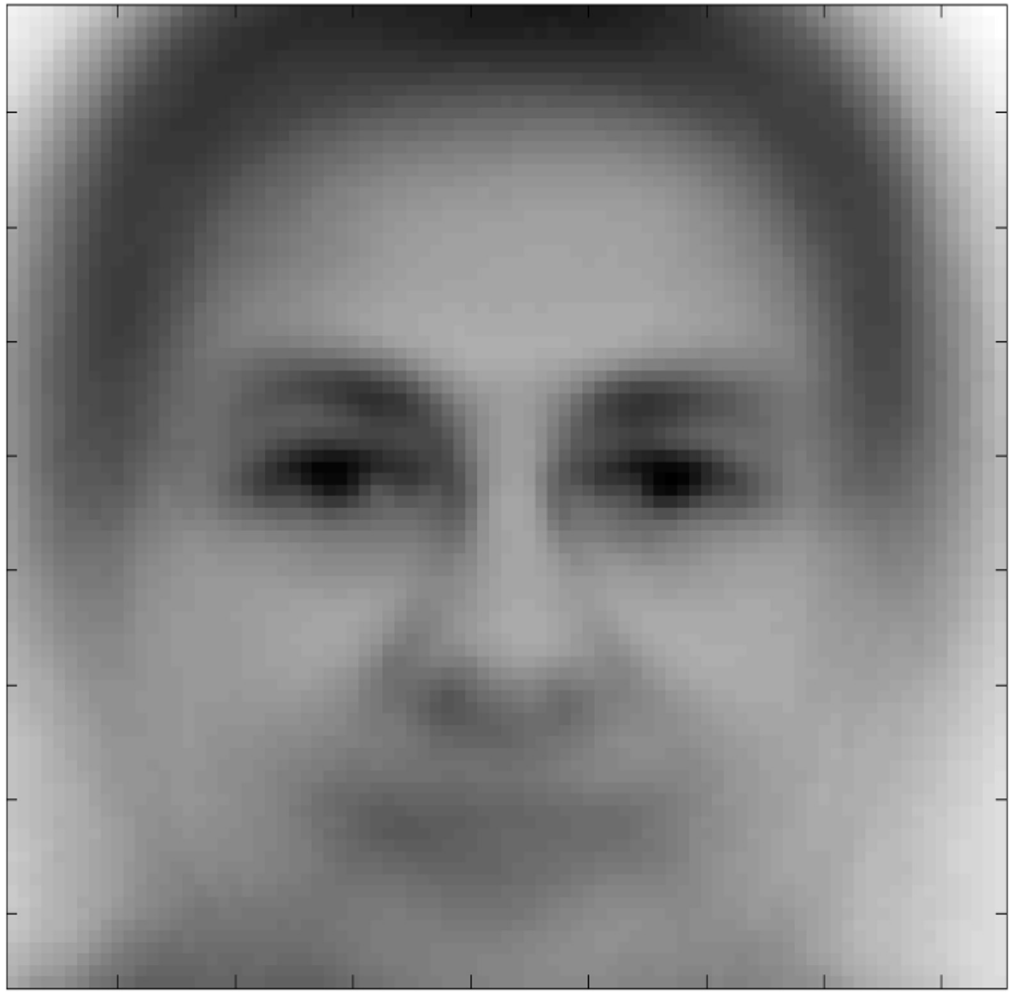


PCA score plots II

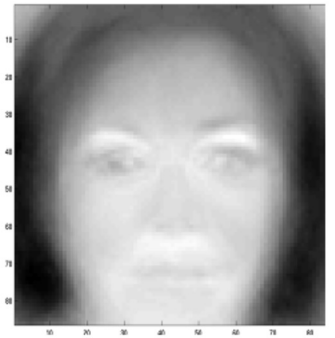
- PCA plot, 1st component is related to gender

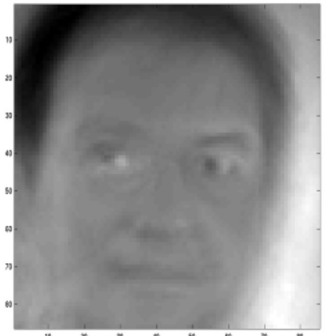








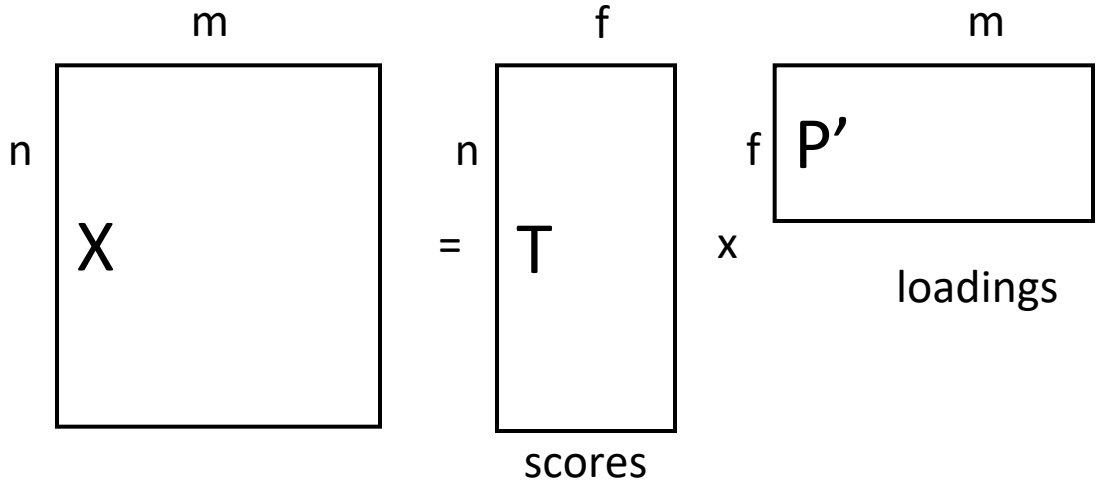


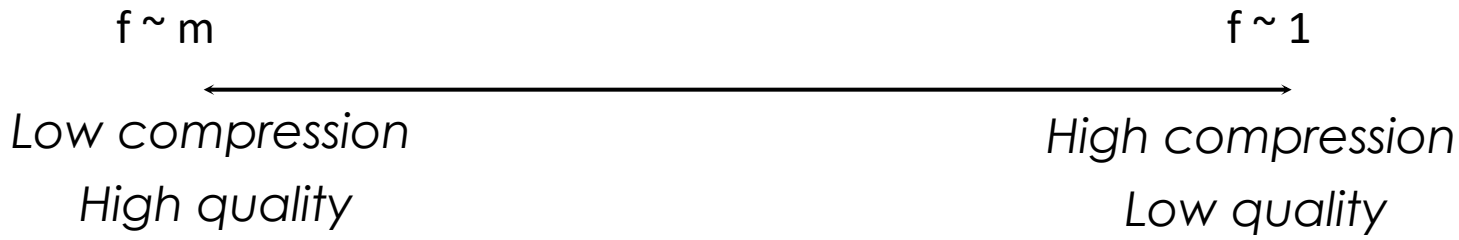


m + A pc1 + B pc2

Data modelling

- PCA is an example of data compression (data coding)
- Helps to understand that dimensionality reduction is not *loosless*





Data modelling



f ~ m

f ~ 1

Low compression

High compression